Run report: itemOJ-run-20261018T2124Z

Descriptive record of one recorded run of examples/n2_routes.rs, mode run (Parts O/J) under `research/weil_descent_angles_20261018/PROTOCOL.md` (Addendum A and Amendments B–G and Addendum H, each committed before the run it governs). It lists what ran, on what host, and what the tool recorded, file by file. Interpretation is in the study's `REPORT.md`. Costs are counted operations (F_p multiplications and inversions, group operations, native operations), never wall-clock; the only times are the tool's own seconds. Isolation: none earned (a shared cloud container, or a RunPod CPU pod whose lscpu is in host.txt); this affects only the seconds, since every recorded value is an exact count or a counted operation total.

env.txt

```
commit 4396b057a1a918da5c98bbbed0b10d5d04776a75
cmd: n2 research/weil_descent_angles_20261018/results/itemOJ-run-20261018T2124Z check; n2 research/weil_descent_angles_20261018/results/itemOJ-run-20261018T2124Z run (N2_THREADS=4, sizes 20..40)
rustc 1.97.0 (2d8144b78 2026-07-07)
4
Sun Oct 18 21:24:50 UTC 2026
post-run (d2edb2ab8): the tool was rustfmt-formatted (format only; unit tests pass); outputs unaffected.
```

tool_log.txt

```
jobs 18 total 57.6s
```

checks.txt

```
K1 decomposition oracle vs brute force (p <= 101): instances 4, points 6068, mismatches 0
K2-K4 p=263 N=69544 h=8 ell=8693: #C(F_p)=264 (p+1=264) | ell·P=O true | tau valid true hom true nonzero true ell·tau(P)=0 true | N·D=0 true
K2-K4 p=271 N=73624 h=8 ell=9203: #C(F_p)=272 (p+1=272) | ell·P=O true | tau valid true hom true nonzero true ell·tau(P)=0 true | N·D=0 true
K2-K4 p=1097 N=1205332 h=4 ell=301333: #C(F_p)=1098 (p+1=1098) | ell·P=O true | tau valid true hom true nonzero true ell·tau(P)=0 true | N·D=0 true
K2-K4 p=1069 N=1142056 h=8 ell=142757: #C(F_p)=1070 (p+1=1070) | ell·P=O true | tau valid true hom true nonzero true ell·tau(P)=0 true | N·D=0 true
K2-K4 p=4229 N=17878244 h=4 ell=4469561: #C(F_p)=4230 (p+1=4230) | ell·P=O true | tau valid true hom true nonzero true ell·tau(P)=0 true | N·D=0 true
K2-K4 p=4507 N=20306164 h=4 ell=5076541: #C(F_p)=4508 (p+1=4508) | ell·P=O true | tau valid true hom true nonzero true ell·tau(P)=0 true | N·D=0 true
K5 p=1123: genus-2 add fast 41.3M+1.01I cantor 112.8M+4.00I | double fast 48.0M+1.00I cantor 115.9M+3.99I | agree true | E(F_p^2) add 12M+1I
K5 p=4561: genus-2 add fast 41.0M+1.00I cantor 113.0M+4.00I | double fast 48.0M+1.00I cantor 116.0M+4.00I | agree true | E(F_p^2) add 12M+1I
K5 p=18251: genus-2 add fast 41.0M+1.00I cantor 113.0M+4.00I | double fast 48.0M+1.00I cantor 116.0M+4.00I | agree true | E(F_p^2) add 12M+1I
K5 p=71963: genus-2 add fast 41.0M+1.00I cantor 113.0M+4.00I | double fast 48.0M+1.00I cantor 116.0M+4.00I | agree true | E(F_p^2) add 12M+1I
```

summary.txt

```
O/J bits 20 arm J c 1: IC/rho ratio in [0.507, 1.542] over 12 cells -> unresolved; steps/p 0.311 (rho 0.338)
O/J bits 20 arm J c 2: IC/rho ratio in [0.508, 1.533] over 12 cells -> unresolved; steps/p 0.311 (rho 0.338)
O/J bits 20 arm J c 4: IC/rho ratio in [0.501, 1.488] over 12 cells -> unresolved; steps/p 0.241 (rho 0.338)
O/J bits 20 arm D c 1: IC/rho ratio in [2.950, 4.183] over 6 cells -> rho cheaper in every cell; steps/p 0.397 (rho 0.338)
O/J bits 20 arm D c 2: IC/rho ratio in [2.489, 3.491] over 6 cells -> rho cheaper in every cell; steps/p 0.320 (rho 0.338)
O/J bits 20 arm D c 4: IC/rho ratio in [3.368, 4.806] over 6 cells -> rho cheaper in every cell; steps/p 0.457 (rho 0.338)
O/J bits 24 arm J c 1: IC/rho ratio in [0.661, 1.954] over 12 cells -> unresolved; steps/p 0.351 (rho 0.283)
O/J bits 24 arm J c 2: IC/rho ratio in [0.680, 1.995] over 12 cells -> unresolved; steps/p 0.364 (rho 0.283)
O/J bits 24 arm J c 4: IC/rho ratio in [0.659, 1.899] over 12 cells -> unresolved; steps/p 0.321 (rho 0.283)
O/J bits 24 arm D c 1: IC/rho ratio in [9.111, 15.079] over 6 cells -> rho cheaper in every cell; steps/p 0.461 (rho 0.283)
O/J bits 24 arm D c 2: IC/rho ratio in [9.781, 16.216] over 6 cells -> rho cheaper in every cell; steps/p 0.498 (rho 0.283)
O/J bits 24 arm D c 4: IC/rho ratio in [6.910, 11.350] over 6 cells -> rho cheaper in every cell; steps/p 0.339 (rho 0.283)
O/J bits 28 arm J c 1: IC/rho ratio in [0.707, 2.119] over 12 cells -> unresolved; steps/p 0.338 (rho 0.425)
O/J bits 28 arm J c 2: IC/rho ratio in [0.681, 2.036] over 12 cells -> unresolved; steps/p 0.320 (rho 0.425)
O/J bits 28 arm J c 4: IC/rho ratio in [0.700, 2.095] over 12 cells -> unresolved; steps/p 0.330 (rho 0.425)
O/J bits 28 arm D c 1: IC/rho ratio in [12.384, 23.683] over 6 cells -> rho cheaper in every cell; steps/p 0.357 (rho 0.425)
O/J bits 28 arm D c 2: IC/rho ratio in [13.013, 24.874] over 6 cells -> rho cheaper in every cell; steps/p 0.376 (rho 0.425)
O/J bits 28 arm D c 4: IC/rho ratio in [13.429, 25.760] over 6 cells -> rho cheaper in every cell; steps/p 0.379 (rho 0.425)
O/J bits 32 arm J c 1: IC/rho ratio in [0.900, 2.774] over 12 cells -> unresolved; steps/p 0.348 (rho 0.378)
O/J bits 32 arm J c 2: IC/rho ratio in [0.701, 2.143] over 12 cells -> unresolved; steps/p 0.262 (rho 0.378)
O/J bits 32 arm J c 4: IC/rho ratio in [0.888, 2.728] over 12 cells -> unresolved; steps/p 0.339 (rho 0.378)
O/J bits 32 arm D c 1: IC/rho ratio in [17.961, 38.797] over 6 cells -> rho cheaper in every cell; steps/p 0.331 (rho 0.378)
O/J bits 32 arm D c 2: IC/rho ratio in [20.006, 43.090] over 6 cells -> rho cheaper in every cell; steps/p 0.371 (rho 0.378)
O/J bits 32 arm D c 4: IC/rho ratio in [20.953, 45.133] over 6 cells -> rho cheaper in every cell; steps/p 0.389 (rho 0.378)
O/J bits 36 arm J c 1: IC/rho ratio in [0.589, 1.852] over 12 cells -> unresolved; steps/p 0.343 (rho 0.626)
O/J bits 36 arm J c 2: IC/rho ratio in [0.596, 1.874] over 12 cells -> unresolved; steps/p 0.347 (rho 0.626)
O/J bits 36 arm J c 4: IC/rho ratio in [0.566, 1.778] over 12 cells -> unresolved; steps/p 0.328 (rho 0.626)
O/J bits 36 arm D c 1: IC/rho ratio in [16.647, 40.044] over 6 cells -> rho cheaper in every cell; steps/p 0.418 (rho 0.626)
O/J bits 36 arm D c 2: IC/rho ratio in [16.535, 39.834] over 6 cells -> rho cheaper in every cell; steps/p 0.411 (rho 0.626)
O/J bits 36 arm D c 4: IC/rho ratio in [15.745, 37.966] over 6 cells -> rho cheaper in every cell; steps/p 0.393 (rho 0.626)
```

```
O/J bits 40 arm J c 1: IC/rho ratio in [0.528, 1.680] over 12 cells -> unresolved; steps/p 0.394 (rho 0.812)
O/J bits 40 arm J c 2: IC/rho ratio in [0.410, 1.301] over 12 cells -> unresolved; steps/p 0.305 (rho 0.812)
O/J bits 40 arm J c 4: IC/rho ratio in [0.370, 1.169] over 12 cells -> unresolved; steps/p 0.274 (rho 0.812)
O/J bits 40 arm D c 1: IC/rho ratio in [12.866, 33.867] over 6 cells -> rho cheaper in every cell; steps/p 0.379 (rho 0.812)
O/J bits 40 arm D c 2: IC/rho ratio in [12.720, 33.332] over 6 cells -> rho cheaper in every cell; steps/p 0.381 (rho 0.812)
O/J bits 40 arm D c 4: IC/rho ratio in [13.185, 34.676] over 6 cells -> rho cheaper in every cell; steps/p 0.391 (rho 0.812)
O/J scaling arm R c 0: cost ~ p^0.787
O/J scaling arm J c 1: cost ~ p^0.792
O/J scaling arm J c 2: cost ~ p^0.762
O/J scaling arm J c 4: cost ~ p^0.758
O/J scaling arm D c 1: cost ~ p^1.072
O/J scaling arm D c 2: cost ~ p^1.086
O/J scaling arm D c 4: cost ~ p^1.076
O/J all answers verified: true
```

ranking.csv (first 12 of 36 rows)

bits	arm	c	cells	ic_cheaper_cells	rho_cheaper_cells	min_ratio	max_ratio	verdict
20	J	1	12	6	6	0.5069	1.5417	unresolved
20	J	2	12	6	6	0.5077	1.5327	unresolved
20	J	4	12	6	6	0.5007	1.4878	unresolved
20	D	1	6	0	6	2.9501	4.1828	rho cheaper in every cell
20	D	2	6	0	6	2.4895	3.4906	rho cheaper in every cell
20	D	4	6	0	6	3.3680	4.8059	rho cheaper in every cell
24	J	1	12	6	6	0.6609	1.9539	unresolved
24	J	2	12	6	6	0.6803	1.9946	unresolved
24	J	4	12	6	6	0.6595	1.8993	unresolved
24	D	1	6	0	6	9.1107	15.0789	rho cheaper in every cell
24	D	2	6	0	6	9.7812	16.2163	rho cheaper in every cell
24	D	4	6	0	6	6.9102	11.3495	rho cheaper in every cell

routes.csv (first 12 of 378 rows)

bits	instance	p	N	h	ell	target	arm	c	factor_base	steps	relations	b_relations	jac_ops	walk_M	walk_I	walk_native	jac_M	jac_I	la_M	la_I	la_native	eadd_M	eadd_I	verified	seconds
20	0	1091	1190984	8	148873	0	R	0	0	291	0	0	0	49105	3270	312	0	0	0	0	0	12	1	1	0.001
20	0	1091	1190984	8	148873	0	J	1	34	281	151	1	1941	89746	2048	942	87753	2025	0	0	1045	12	1	1	0.001
20	0	1091	1190984	8	148873	0	D	1	34	607	212	2	0	322779	7494	2928	0	0	0	0	1594	12	1	1	0.003
20	0	1091	1190984	8	148873	0	J	2	67	184	114	4	1755	81811	1860	724	79867	1837	0	0	688	12	1	1	0.001
20	0	1091	1190984	8	148873	0	D	2	67	381	166	6	0	214174	5420	2289	0	0	0	0	1135	12	1	1	0.002
20	0	1091	1190984	8	148873	0	J	4	133	200	104	6	1812	84998	1910	875	82228	1887	0	0	560	12	1	1	0.001
20	0	1091	1190984	8	148873	0	D	4	133	369	149	13	0	212847	5377	2372	0	0	0	0	815	12	1	1	0.003
20	0	1091	1190984	8	148873	1	R	0	0	317	0	0	0	49931	3318	358	0	0	0	0	0	12	1	1	0.001
20	0	1091	1190984	8	148873	1	J	1	34	356	182	3	1948	87715	1934	1146	85497	1911	0	0	1264	12	1	1	0.001
20	0	1091	1190984	8	148873	1	D	1	34	259	92	1	0	151137	4086	1345	0	0	0	0	552	12	1	1	0.001
20	0	1091	1190984	8	148873	1	J	2	67	244	123	2	1804	81555	1787	875	79370	1764	0	0	752	12	1	1	0.001
20	0	1091	1190984	8	148873	1	D	2	67	251	99	3	0	150811	4114	1519	0	0	0	0	622	12	1	1	0.001

routes_summary.csv (first 12 of 324 rows)

bits	arm	c	ipr	jac_pricing	native_w	runs	verified	mean_p	mean_cost	mean_rho_cost	ratio_to_rho	mean_steps_over_p	rho_steps_over_p
20	J	1	3M	measured	0	9	1	1081	94779.2	62581.1	1.5145	0.3112	0.3379
20	J	1	3M	measured	0.1	9	1	1081	95011.3	62621.4	1.5172	0.3112	0.3379
20	J	1	3M	measured	1	9	1	1081	97099.9	62984.0	1.5417	0.3112	0.3379
20	J	1	3M	eadd	0	9	1	1081	31725.0	62581.1	0.5069	0.3112	0.3379
20	J	1	3M	eadd	0.1	9	1	1081	31957.1	62621.4	0.5103	0.3112	0.3379
20	J	1	3M	eadd	1	9	1	1081	34045.7	62984.0	0.5405	0.3112	0.3379
20	J	1	2log2p_M	measured	0	9	1	1081	128568.6	122380.8	1.0506	0.3112	0.3379
20	J	1	2log2p_M	measured	0.1	9	1	1081	128800.7	122421.1	1.0521	0.3112	0.3379
20	J	1	2log2p_M	measured	1	9	1	1081	130889.3	122783.7	1.0660	0.3112	0.3379
20	J	1	2log2p_M	eadd	0	9	1	1081	65908.6	122380.8	0.5386	0.3112	0.3379
20	J	1	2log2p_M	eadd	0.1	9	1	1081	66140.7	122421.1	0.5403	0.3112	0.3379
20	J	1	2log2p_M	eadd	1	9	1	1081	68229.3	122783.7	0.5557	0.3112	0.3379

Graph

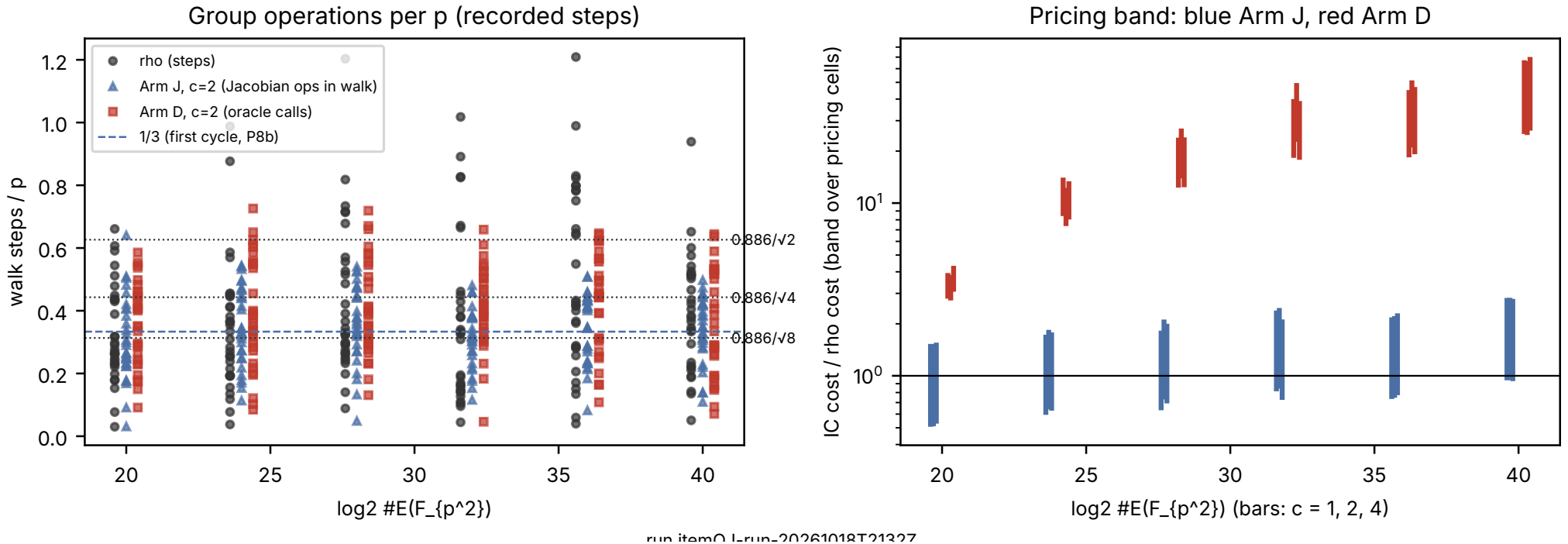


Figure 1: Graph drawn from the recorded re-run 20261018T2132Z (this run's rho rows are superseded by Amendment G); steps/p and pricing bands.